Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **The Mitra Team**
b. Team leader name: **Gaurav Giri**
c. Problem Statement: Media, Content & Digital Experiences. Correct explainer videos are slow and costly to make, and AI video gets facts wrong.
d. Team members: Gaurav Giri (leader), Viraj Sawad

Brief about the idea

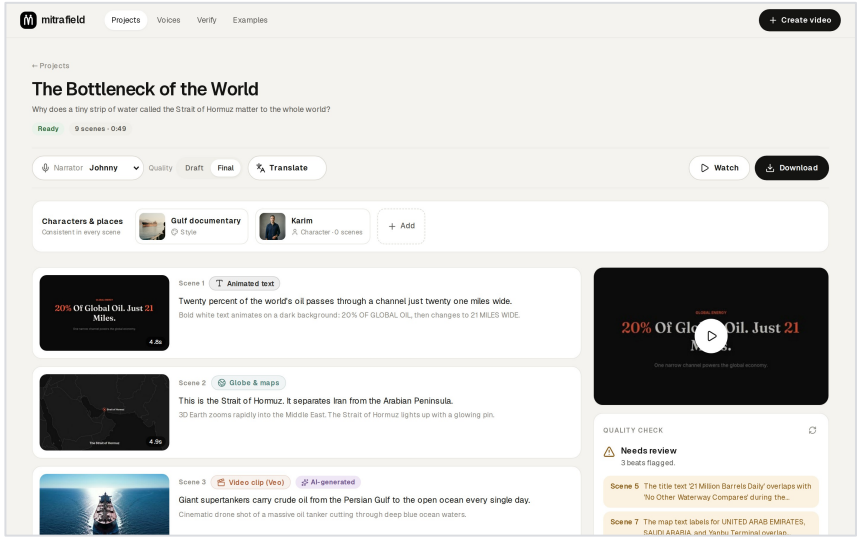
Ask a question. Get a correct, narrated explainer video.

MitraField is an AI video studio. An AI director built on Gemini writes the script, chooses the right kind of picture for every scene and renders a finished video with narration, music and subtitles, in minutes.

- Real data for facts
- Your own voice
- 9 languages

Problem: a correct explainer takes a studio days; AI video tools are fast but invent numbers, maps and equations; most explainers are English-only.

For: teachers, NGOs, newsrooms and creators across JAPAC.



The studio: approve the storyboard, edit any scene, AI pictures labelled

Opportunities

How is it different from existing ideas?

- Picks the right kind of picture per scene: real maps, math animation, charts; AI imagery only for mood
- Full narrated explainers of 1 to 6 minutes, not 8-second clips
- Checks its own work: facts, picture and sound

How will it solve the problem?

- Minutes instead of days, for about $1 to $3 a video
- Facts drawn from data and code, then fact-checked with Google Search
- One click makes a Hindi, Nepali or Bahasa version with the same visuals

USP

- Correct, not just pretty
- Narrated in your own cloned voice
- An invisible watermark anyone can verify
- You approve the plan first, then change any scene in plain words

Beautiful is easy. Correct is the product.

List of features offered by the solution

1 AI director

Writes the storyboard from a question, notes or PDFs

2 Eleven kinds of visual

Real maps, math animation, charts, timelines, Veo clips

3 Voice cloning

Your voice from 15 seconds of speech, in English and Hindi

4 One-click translation

Nine narration languages; the visuals are reused

5 Edit in plain words

"Simpler", "show it on a map": only that scene re-renders

6 Fact and quality checks

Google Search grounding; Gemini reviews every scene

7 Ten looks, one cast

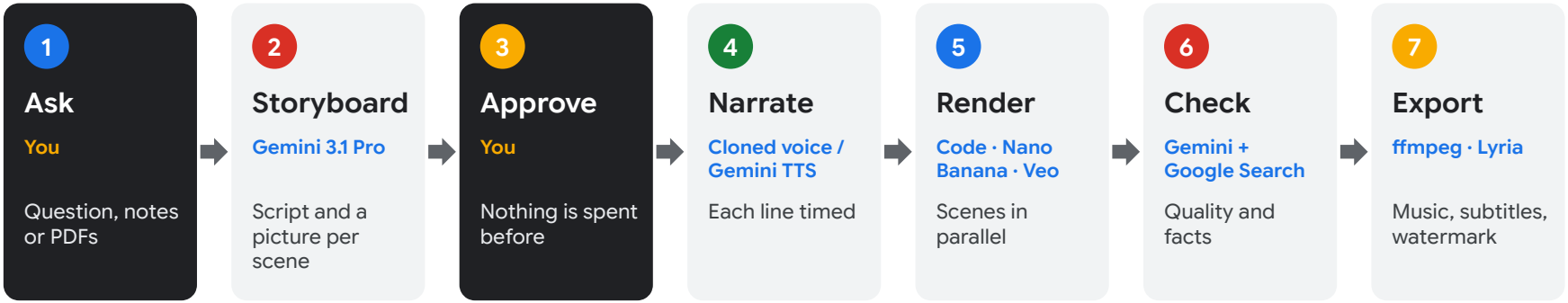
The same characters and objects in every shot

8 Invisible watermark

Anyone can verify a copy on the Verify page

Also: vertical Shorts, subtitles, a music bed, cost per video, spending caps, and it works on a phone.

Process flow diagram or Use-case diagram



Change one scene

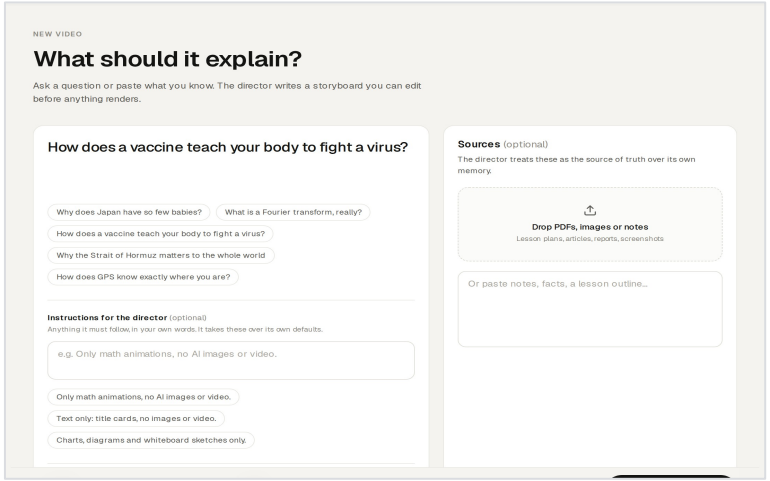
Ask in plain words → Only that scene re-renders → Update video

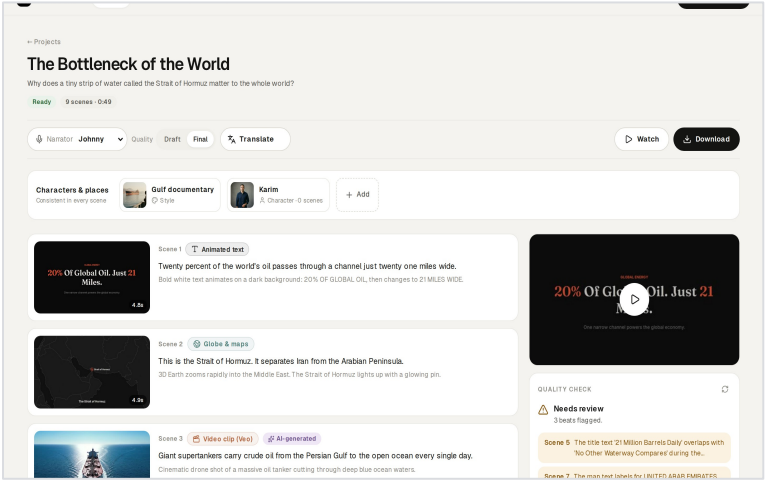
Translate

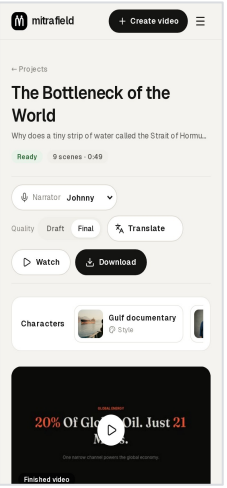
Pick a language → New narration, same visuals → Hindi / Nepali version

Every scene is narrated first, so its picture is cut to the exact length of the sentence.

Wireframes/Mock diagrams of the proposed solution (optional)







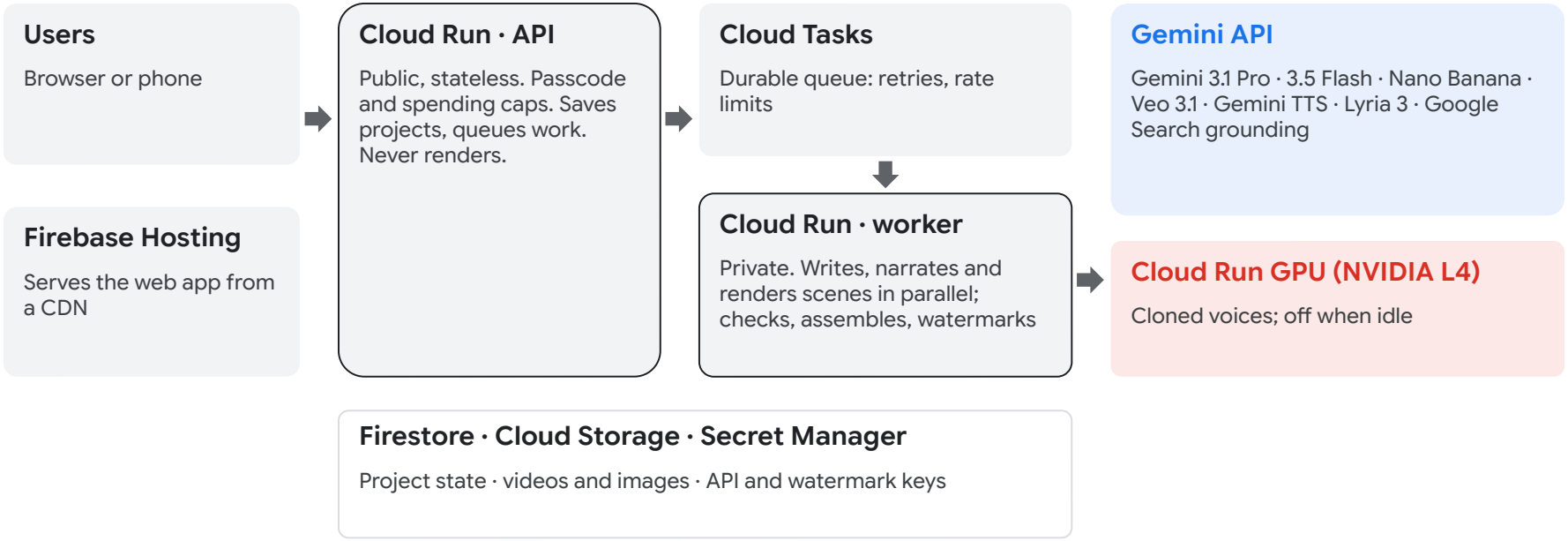
1. Create: question, sources, instructions, narrator and look. The main action stays in view.

2. Studio: approve the storyboard, edit any scene, render, download or translate.

3. On a phone

Screens from the working prototype, not mock-ups.

Architecture diagram of the proposed solution



Delivery GitHub Actions → keyless sign-in → Cloud Build → Artifact Registry → test copy checked live → released

Technologies to be used in the solution

Google AI

- Gemini 3.1 Pro
- Gemini 3.5 Flash
- Nano Banana
- Veo 3.1
- Gemini TTS
- Lyria 3
- Google Search grounding
- SynthID (on generated media)

Google Cloud

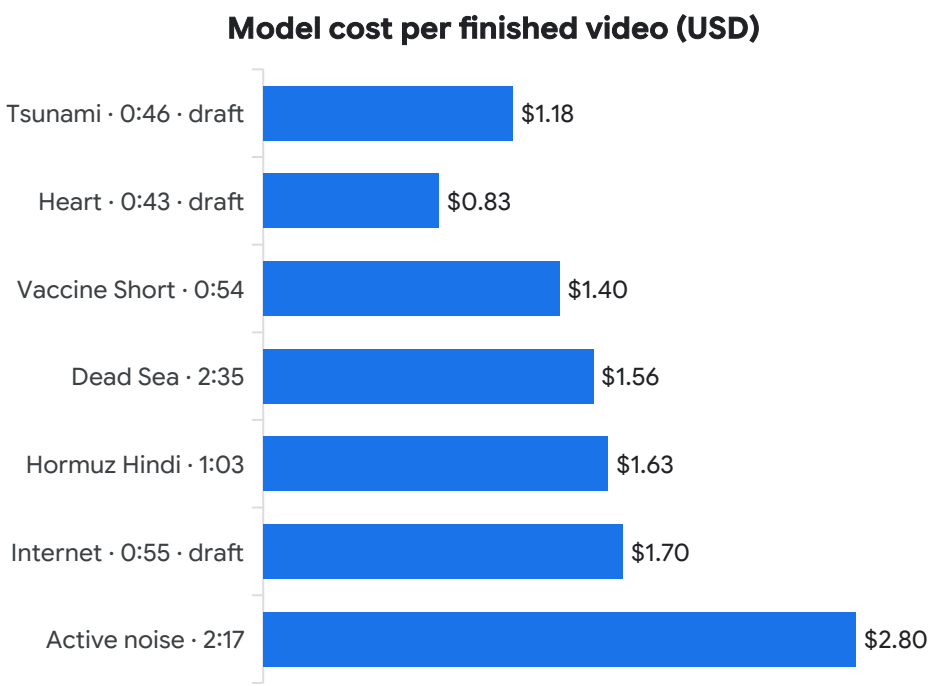
- Cloud Run
- Cloud Run GPU (L4)
- Cloud Tasks
- Firestore
- Cloud Storage
- Secret Manager
- Firebase Hosting
- Cloud Build
- Artifact Registry
- Workload Identity Federation
- Cloud Billing budgets

Open source

- FastAPI
- Next.js
- Manim
- ffmpeg
- OmniVoice
- Natural Earth
- Headless Chrome
- OpenCV

Everything runs on Google Cloud in asia-southeast1 (Singapore); all AI calls go through the Gemini API.

Estimated implementation cost (optional)



$0.80–2.80

model cost per finished video, logged per call

$0

compute when idle: every service scales to zero

42% · 31% · 24%

Veo clips · images · Gemini text (share of spend)

$5 / $25

hard caps per video and per day

Estimates from Google's published Gemini API rates, logged by the app for every call. Cloned-voice GPU time is billed only while narrating (about $1 an hour when on, our estimate).

Snapshots of the prototype



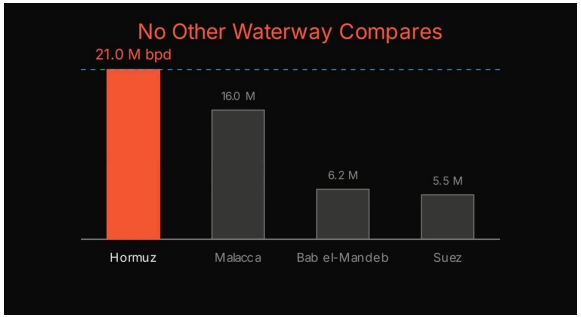
Woodblock look · how a tsunami forms



Vertical Short · vaccines, with a presenter



Hindi version · labels redrawn in Devanagari



Math animation, drawn by code



Real map data · the Strait of Hormuz



Veo clip from an on-model first frame

Prototype Performance report/Benchmarking

12
finished videos, 116 scenes, 3 languages

62%
of scenes drawn from data and code

4–9 min
to render a one-minute video, unattended

126
automated tests, run on every change

Measure	Result
Cloned narration (Cloud Run GPU)	18 to 46 s for the first line after idle (GPU start); 0.7 to 6 s per line when warm
Fact check (Hormuz video)	All 9 checkable claims supported, 12 web sources
Watermark robustness	Read back after re-encoding (crf 34), 480p, 30 to 24 fps and a trimmed start; ~7 s to verify a 46 s video
Release safety	Each release is a test copy first; goes live only after a live check of the database, a model call, the watermark and the voice GPU

Responsible AI and trust

Fact check

Every claim checked against your sources, then Google Search, with links

AI labels

Scenes with AI pictures are labelled; Veo and Nano Banana output carries SynthID

Invisible watermark

In every export; anyone can verify a copy on the Verify page

No deepfakes

Requests to fake a real person's words are refused

Safe with uploads

PDFs and notes are treated as data; model-written code runs in a sandbox

Spending limits

Every call priced and logged; caps of $5 a video and $25 a day

Consent is required before a voice is cloned, and uploaded sources and voice samples are never public.

Additional Details/Future Development (if any)

1. **Google Drive**
   Import sources from Drive and save finished videos to it
2. **A fully flexible video editor**
   Timeline editing on top of the AI storyboard
3. **Sign-in and private workspaces**
   Firebase Authentication, for schools and teams
4. **More languages and voices**
   Regional voices and subject glossaries
5. **Publish straight to YouTube**
   And to learning platforms

Known limits today

- No user accounts yet; creating needs a shared passcode
- The first cloned-voice line after idle waits for the GPU to start
- Fact checks reduce errors but cannot remove them all; flagged claims are shown
- Heavy cropping or editing can remove the watermark

Provide links to your:

GitHub public repository
github.com/ViraiSawad1021/MitraField

Demo video (2:08)
github.com/ViraiSawad1021/MitraField/blob/main/docs/demo-video/MitraField_Demo.mp4

Final product
mitrafield-api-u6zih6e3hq-as.a.run.app

Browse the examples freely. Creating a video needs the passcode given in our submission form.

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Thank you

